

Database Management Systems

Lab – 2

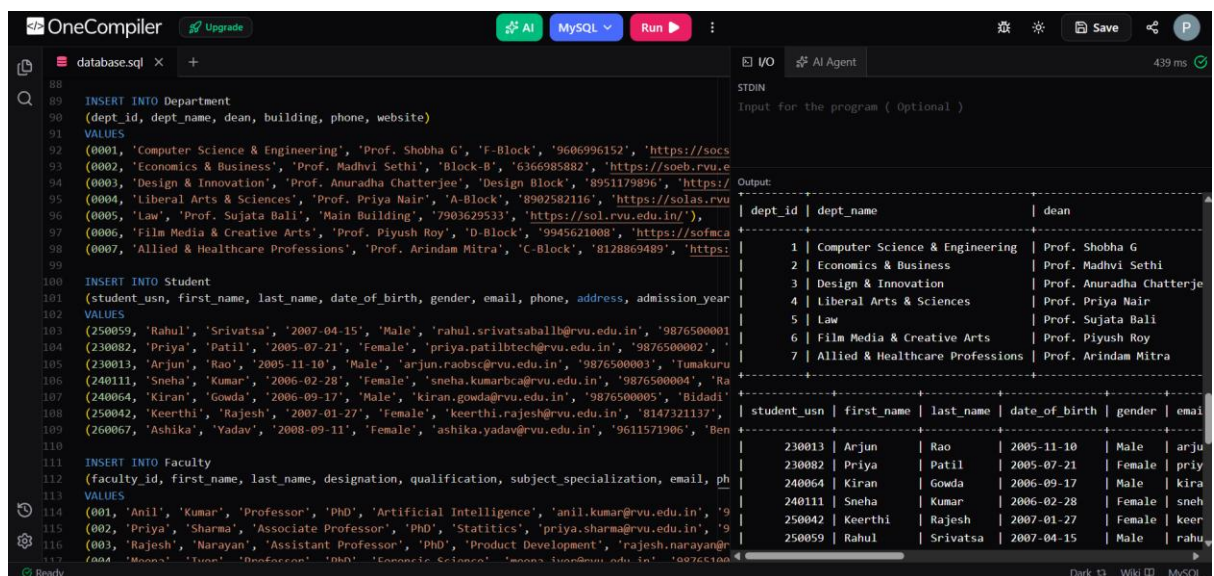
▪ Data Manipulation Language (DML) Commands:

1. INSERT

2. UPDATE

3. DELETE

➤ INSERT –



The screenshot shows the OneCompiler MySQL interface. The left pane contains SQL code for inserting data into three tables: Department, Student, and Faculty. The right pane shows the output of the queries, which are formatted as tables.

```
88
89 INSERT INTO Department
90 (dept_id, dept_name, dean, building, phone, website)
91 VALUES
92 (0001, 'Computer Science & Engineering', 'Prof. Shobha G', 'F-Block', '9606996152', 'https://socs
93 (0002, 'Economics & Business', 'Prof. Madhvi Sethi', 'Block-B', '6366985882', 'https://soeb.rvu.e
94 (0003, 'Design & Innovation', 'Prof. Anuradha Chatterjee', 'Design Block', '8951179896', 'https:/
95 (0004, 'Liberal Arts & Sciences', 'Prof. Priya Nair', 'A-Block', '8902582116', 'https://solas.rvu
96 (0005, 'Law', 'Prof. Sujata Bali', 'Main Building', '7903629533', 'https://sol.rvu.edu.in/'),
97 (0006, 'Film Media & Creative Arts', 'Prof. Piyush Roy', 'D-Block', '9945621008', 'https://sofmca
98 (0007, 'Allied & Healthcare Professions', 'Prof. Arindam Mitra', 'C-Block', '8128869489', 'https:
99
100 INSERT INTO Student
101 (student_usn, first_name, last_name, date_of_birth, gender, email, phone, address, admission_year
102 VALUES
103 (250059, 'Rahul', 'Srivatsa', '2007-04-15', 'Male', 'rahul.srivatsaballb@rvu.edu.in', '9876500001
104 (230082, 'Priya', 'Patil', '2005-07-21', 'Female', 'priya.patilbtech@rvu.edu.in', '9876500002', '
105 (230013, 'Arjun', 'Rao', '2005-11-10', 'Male', 'arjun.raobsc@rvu.edu.in', '9876500003', 'Tumakuru
106 (240111, 'Sneha', 'Kumar', '2006-02-28', 'Female', 'sneha.kumarbca@rvu.edu.in', '9876500004', 'Ra
107 (240064, 'Kiran', 'Gowda', '2006-09-17', 'Male', 'kiran.gowda@rvu.edu.in', '9876500005', 'Bidadi
108 (250042, 'Keerthi', 'Rajesh', '2007-01-27', 'Female', 'keerthi.rajesh@rvu.edu.in', '8147321137',
109 (260067, 'Ashika', 'Yadav', '2008-09-11', 'Female', 'ashika.yadav@rvu.edu.in', '9611571906', 'Ben
110
111 INSERT INTO Faculty
112 (faculty_id, first_name, last_name, designation, qualification, subject_specialization, email, ph
113 VALUES
114 (001, 'Anil', 'Kumar', 'Professor', 'PhD', 'Artificial Intelligence', 'anil.kumar@rvu.edu.in', '9
115 (002, 'Priya', 'Sharma', 'Associate Professor', 'PhD', 'Statistics', 'priya.sharma@rvu.edu.in', '9
116 (003, 'Rajesh', 'Narayan', 'Assistant Professor', 'PhD', 'Product Development', 'rajesh.narayan@r
117 (004, 'Ramesh', 'Taneja', 'Associate Professor', 'PhD', 'Enterprise Systems', 'ramesh.taneja@rvu.edu.in', '00765100
```

Output:

dept_id	dept_name	dean
1	Computer Science & Engineering	Prof. Shobha G
2	Economics & Business	Prof. Madhvi Sethi
3	Design & Innovation	Prof. Anuradha Chatterjee
4	Liberal Arts & Sciences	Prof. Priya Nair
5	Law	Prof. Sujata Bali
6	Film Media & Creative Arts	Prof. Piyush Roy
7	Allied & Healthcare Professions	Prof. Arindam Mitra

student_usn	first_name	last_name	date_of_birth	gender	email
230013	Arjun	Rao	2005-11-10	Male	arju
230082	Priya	Patil	2005-07-21	Female	priy
240064	Kiran	Gowda	2006-09-17	Male	kira
240111	Sneha	Kumar	2006-02-28	Female	sneh
250042	Keerthi	Rajesh	2007-01-27	Female	keer
250059	Rahul	Srivatsa	2007-04-15	Male	rahu

faculty_id	first_name	last_name	designation	qualification	subject_specialization	email	ph
001	Anil	Kumar	Professor	PhD	Artificial Intelligence	anil.kumar@rvu.edu.in	9
002	Priya	Sharma	Associate Professor	PhD	Statistics	priya.sharma@rvu.edu.in	9
003	Rajesh	Narayan	Assistant Professor	PhD	Product Development	rajesh.narayan@rvu.edu.in	9


```

150 SELECT * FROM Department;
151 SELECT * FROM Student;
152 SELECT * FROM Faculty;
153 SELECT * FROM Courses;
154 SELECT * FROM Enrollment;

```

➤ UPDATE –

The screenshot shows the OneCompiler IDE with a MySQL database. The left pane displays a series of SQL update queries (lines 144-172) for the Department, Student, Faculty, Courses, and Enrollment tables. The right pane shows the output of these queries, which are formatted as tables.

Department Table Output:

dept_id	dept_name	dean	building
1	Computer Science & Engineering	Prof. Vikram Reddy	New Tech Bloc
2	Economics & Business	Prof. Madhvi Sethi	Block-B
3	Design & Innovation	Prof. Anuradha Chatterjee	Design Block
4	Liberal Arts & Sciences	Prof. Priya Nair	A-Block
5	Law	Prof. Sujata Bali	Main Building
6	Film Media & Creative Arts	Prof. Piyush Roy	D-Block
7	Allied & Healthcare Professions	Prof. Arindam Mitra	C-Block

Student Table Output:

student_usn	first_name	last_name	date_of_birth	gender	email
230013	Arjun	Rao	2005-11-10	Male	arjun.raobsc@rvu.edu.
230082	Priya	Patil	2005-07-21	Female	priya.patilbtech@rvu.
240064	Kiran	Gowda	2006-09-17	Male	kiran.gowda@rvu.edu.i
240111	Sneha	Kumar	2006-02-28	Female	sneha.kumarbca@rvu.ed
250042	Keerthi	Rajesh	2007-01-27	Female	keerthi.rajesh@rvu.ed
250059	Rahul	Srivatsa	2007-04-15	Male	rahul.srivatsaballb@

Output:

dept_id	dept_name	dean	building	phone	website
1	Computer Science & Engineering	Prof. Vikram Reddy	New Tech Block	9606996152	https://socse.rvu.edu.in/
2	Economics & Business	Prof. Madhvi Sethi	Block-B	9999999999	https://soeb.rvu.edu.in/
3	Design & Innovation	Prof. Anuradha Chatterjee	Design Block	8951179896	https://newsite.rvu.edu.in/
4	Liberal Arts & Sciences	Prof. Priya Nair	A-Block	8902582116	https://solas.rvu.edu.in/
5	Law	Prof. Sujata Bali	Main Building	7903629533	https://sol.rvu.edu.in/
6	Film Media & Creative Arts	Prof. Piyush Roy	D-Block	9945621008	https://sofmc.rvu.edu.in/
7	Allied & Healthcare Professions	Prof. Arindam Mitra	C-Block	8128869489	https://soahp.rvu.edu.in/

student_usn	first_name	last_name	date_of_birth	gender	email	phone	address	admission_year
230013	Arjun	Rao	2005-11-10	Male	arjun.raobsc@rvu.edu.in	9876500003	Tumakuru	2023
230082	Priya	Patil	2005-07-21	Female	priya.patilbtech@rvu.edu.in	9876500002	Mysuru	2023
240064	Kiran	Gowda	2006-09-17	Male	kiran.gowda@rvu.edu.in	9876500005	Bidadi	2024
240111	Sneha	Kumar	2006-02-28	Female	sneha.kumarbca@rvu.edu.in	9876500004	Ramanagara	2024
250042	Keerthi	Rajesh	2007-01-27	Female	keerthi.rajesh@rvu.edu.in	8147321137	Bengaluru	2024
250059	Rahul	Srivatsa	2007-04-15	Male	rahul.srivatsaballb@rvu.edu.in	9876500010	Whitefield, Bengaluru	2024

Output:

260067	Ashika	Yadav	2008-09-11	Female	ashika.yadav@rvu.edu.in	9611571906	Bengaluru	2026
--------	--------	-------	------------	--------	-------------------------	------------	-----------	------

faculty_id	first_name	last_name	designation	qualification	subject_specialization	email	phone	dept
1	Anil	Kumar	Senior Professor	PhD	Artificial Intelligence	anil.kumar@rvu.edu.in	9876510001	
2	Priya	Sharma	Associate Professor	PhD	Statistics	priya.sharma@rvu.edu.in	8765432101	
3	Rajesh	Narayan	Assistant Professor	PhD	Product Development	rajesh.narayan@rvu.edu.in	9876510003	
4	Meena	Iyer	Professor	PhD	Forensic Science	meena.iyer@rvu.edu.in	9876510004	
5	Suresh	Shetty	Assistant Professor	M.Tech	Embedded Systems	suresh.shetty@rvu.edu.in	9876510005	
6	Deepak	Singh	Associate Professor	PhD	Economics	deepak.singh@rvu.edu.in	9876510006	
7	Divya	Nair	Assistant Professor	MA	English Literature	divya.nair@rvu.edu.in	9876510007	

course_id	course_code	course_name	course_type	credits	semester	faculty_id	schedule
1	101	Machine Learning & AI	Core	4	5	1	Mon 09:00AM-10:00AM
2	102	Applied Statistics	Core	3	3	2	Tue 09:00AM-10:00AM
3	103	Project Management	Core	3	5	3	Thu 12:00PM-01:00PM
4	104	Forensic Psychology	Minor	3	5	4	Wed 02:00PM-05:00PM

Output:

1	101	Machine Learning & AI	Core	4	5	1	Mon 09:00AM-10:00AM
2	102	Applied Statistics	Core	3	3	2	Tue 09:00AM-10:00AM
3	103	Project Management	Core	3	5	3	Thu 12:00PM-01:00PM
4	104	Forensic Psychology	Minor	3	5	4	Wed 02:00PM-05:00PM
5	105	Internet of Things	Elective	4	3	5	Fri 12:00PM-01:00PM
6	106	Macroeconomics	Core	4	6	6	Mon 11:00AM-12:00PM
7	107	Philosophy	Elective	3	3	7	Wed 2:00PM-4:00PM

enrollment_id	enrollment_date	student_usn	course_id	academic_year
1	2025-01-15	230013	1	2024
2	2025-01-15	230082	2	2023
3	2025-01-15	240064	3	2024
4	2025-01-15	240111	4	2024
5	2025-01-15	250042	5	2025
6	2025-02-01	250059	6	2025
7	2025-01-15	260067	7	2026

➤ DELETE –

```

166
167 DELETE FROM Enrollment WHERE enrollment_id = 7;
168 DELETE FROM Enrollment WHERE student_usn = 260067;
169 DELETE FROM Enrollment WHERE course_id = 5;
170
171 DELETE FROM Student WHERE student_usn = 250042;
172 DELETE FROM Student WHERE student_usn = 260067;
173

```

enrollment_id	enrollment_date	student_usn	course_id	academic_year
1	2025-01-15	230013	1	2024
2	2025-01-15	230082	2	2023
3	2025-01-15	240064	3	2024
4	2025-01-15	240111	4	2024
6	2025-02-01	250059	6	2025

student_usn	first_name	last_name	date_of_birth	gender	email	phone	address	admission_year	semester
230013	Arjun	Rao	2005-11-10	Male	arjun.raobsc@rvu.edu.in	9876500003	Tumakuru	2023	7
230082	Priya	Patil	2005-07-21	Female	priya.patilbtech@rvu.edu.in	9876500002	Mysuru	2023	7
240064	Kiran	Gowda	2006-09-17	Male	kiran.gowda@rvu.edu.in	9876500005	Bidadi	2024	5
240111	Sneha	Kumar	2006-02-28	Female	sneha.kumarbca@rvu.edu.in	9876500004	Ramanagara	2024	5
250059	Rahul	Srivatsa	2007-04-15	Male	rahul.srivatsaballb@rvu.edu.in	9876500010	Whitefield, Bengaluru	2025	3

semester	section	blood_group
7	B	B+
7	B	A+
5	A	AB+
5	B	O-
3	A	O+